

1) If there is vitamin deficiency, which diseases occur?

- Vitamin A deficiency →
- Vitamin C deficiency →
- Vitamin D deficiency →
- Vitamin K deficiency →
- Vitamin E deficiency →

2) Fill the blanks.

| Organic Molecule | Small Unit (Monomer) | Bond Between Units |
|------------------|----------------------|--------------------|
| Carbohydrates    |                      |                    |
| Proteins         |                      |                    |
| Lipids           |                      |                    |
| Nucleic acids    |                      |                    |

3) In organic molecules, while big molecules are formed, we mentioned **dehydration=condensation** reactions. And while big molecules are disrupted to form little molecules, we called this reactions **hydrolysis**.

Give example for each.

4) Compare fatty acids in the table:

| Feature                   | Saturated Fatty Acids | Unsaturated Fatty Acids |
|---------------------------|-----------------------|-------------------------|
| Bonds                     |                       |                         |
| Shape                     |                       |                         |
| State at Room Temperature |                       |                         |
| Source                    |                       |                         |
| Health Effect             |                       |                         |
| Example                   |                       |                         |

5) Write details about polysaccharides:

Glycogen:

Starch:

Cellulose:

Chitin:

6) In a eukaryotic cell, ribosomes found in multiple locations inside of the cell. More specifically, ribosomes are found **free in the cytoplasm**, **on the nuclear membrane** (membrane of nucleus), and **inside mitochondria, chloroplasts** and **rough endoplasmic reticulum**.

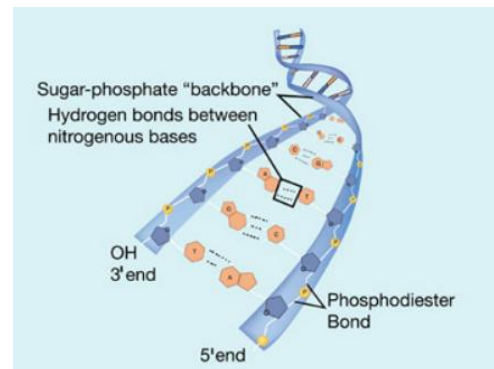
According to this pick two of them and explain why they have their own ribosomes?

1)

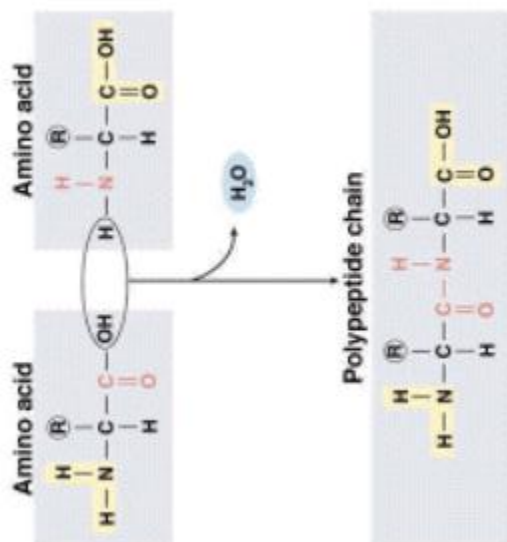
- Don't forget, vitamins are **all essential** nutrients. It means we **cannot produce** them but we have to take them from outside!

2)

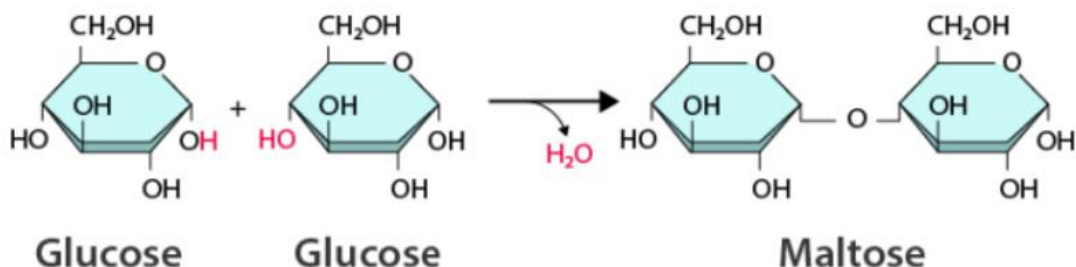
Don't forget, in DNA,  
between nucleotides, there is Phosphodiester bond.  
Between nitrogenous organic bases,  
there are hydrogen bonds.



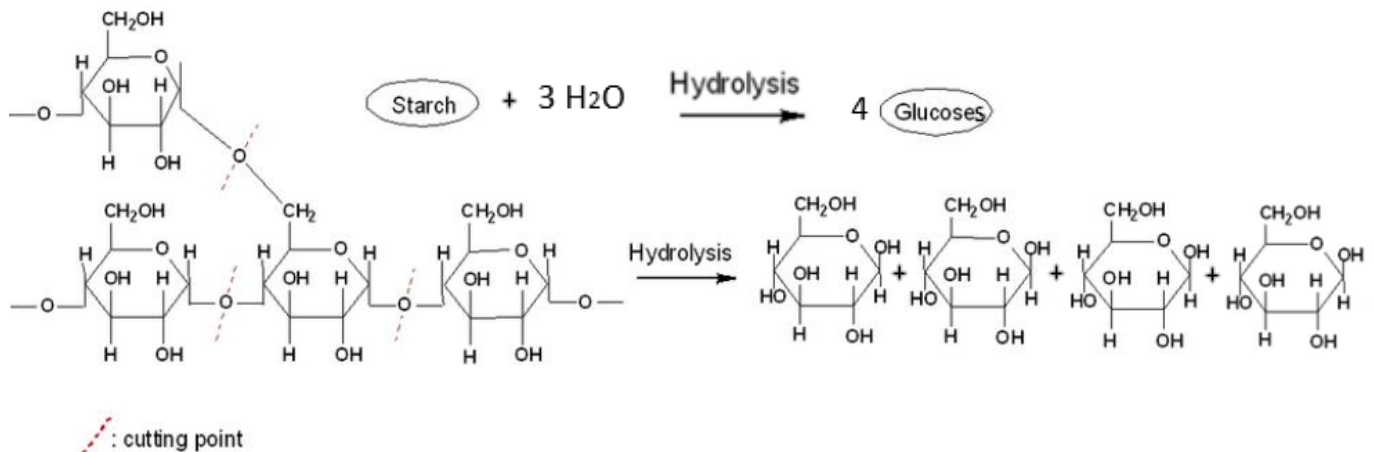
- 3) Peptide bond is formed here in a dehydration reaction. Since two molecules of small units entered the reaction,  $2 - 1 = 1$  bond is formed and 1 molecule of water is produced. Energy is used to combine these two amino acids.



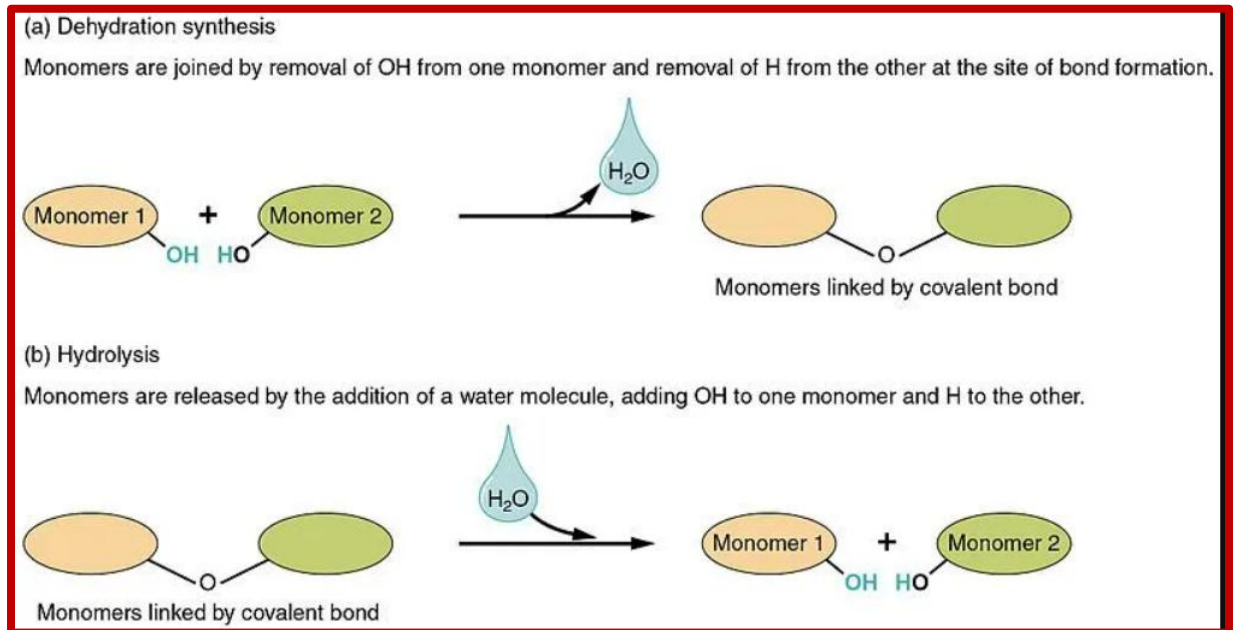
Similarly maltose formation:



Lets check hydrolysis example this time :

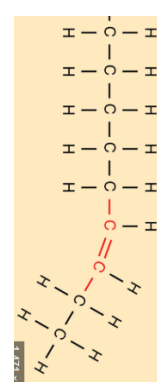
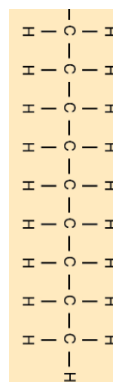


Schematic hydrolysis of starch by amylase enzyme



4)

| Feature                   | Saturated Fatty Acids           | Unsaturated Fatty Acids           |
|---------------------------|---------------------------------|-----------------------------------|
| Bonds                     | Only single bonds               | One or more double bonds          |
| Shape                     | Straight chain                  | Bent chain                        |
| State at Room Temperature | Usually solid                   | Usually liquid                    |
| Source                    | Animal based                    | Plant based                       |
| Health Effect             | Can increase heart disease risk | Generally healthier for the heart |
| Example                   | Butter, animal fat              | Olive oil                         |



5)

| Polysaccharide | Details                                                                                                                             |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Glycogen       | Animal, fungi storage carbohydrate. Human stores them in the liver and muscles.                                                     |
| Starch         | Plant storage carbohydrate. Found in potatoes, rice, and corn. When stained with iodine (Lugol) test, gives dark blue-purple color. |
| Cellulose      | Structure carbohydrate. Makes up plant cell walls. Gives support and strength to plants. Humans cannot digest it.                   |
| Chitin         | Structure carbohydrate. Found in the exoskeleton of insects and the cell walls of fungi. Provides protection and support.           |

- 6) Mitochondria and chloroplasts have their own ribosomes because they can produce some of their own proteins independently inside the organelle. This supports the endosymbiotic theory, which suggests that mitochondria and chloroplasts were once free-living prokaryotic cells.

7)

| Organelle / Structure | Prokaryotic Cell | Eukaryotic Cell |
|-----------------------|------------------|-----------------|
| Nucleus               | No               | Yes             |
| Ribosome              | Yes              | Yes             |
| Mitochondria          | No               | Yes             |
| Chloroplast           | No               | Some have       |
| Endoplasmic Reticulum | No               | Yes             |
| Golgi Apparatus       | No               | Yes             |
| Cell Membrane         | Yes              | Yes             |
| Cytoplasm             | Yes              | Yes             |
| Cell Wall             | Usually yes      | Some have       |
| DNA                   | Yes              | Yes             |

**Note:**

Prokaryotic cells include the kingdoms Bacteria and Archaea.

Eukaryotic cells include the kingdoms Protista, Fungi, Plantae, and Animalia.

Prokaryotic cells do not have membrane-bound organelles because they are simpler and smaller cells without an internal membrane system.

Both prokaryotic and eukaryotic cells always have ribosomes, cell membrane, cytoplasm, and DNA.